



Al-Balqa' Applied University
Faculty of Engineering Technology - Department of Basic Scientific Sciences
Mid Exam - Linear Algebra

Name: أحمد محمد العبدالله

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Part A: Multiple choice

(12 marks)

-Be sure that you have 6 different questions.

-Complete the following table by putting (the correct answer letter) in the table.

Q1	Q2	Q3	Q4	Q5	Q6
b	a	c	b	a	d

Q1) Suppose that the augmented matrix for a system of linear equations has been reduced by row operation to

the given form $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 2 & 2 \end{bmatrix}$ The solution of the linear system (respectively x,y,z) is

- a) -4,3,-2 b) 1, 0, 1 c) -1,0,-1 d) 4, -3, 2

$$x + y + 7z = -7$$

Q2) Determine the value of a for which the system $2x + 3y + 17z = 11$ has no solution

$$x + 2y + (a^2 + 1)z = 6a$$

- a) -3 b) -2 c) 2 d) 3

Q3: If $(2I + 4A)^{-1} = \begin{bmatrix} \frac{1}{6} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{10} \end{bmatrix}$ then $\text{tr}(A) =$

- a) 4 b) 14 c) 3 d) 6

Q4: Let the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 5 & 4 \\ -1 & 2 & 6 \end{bmatrix}$ the cofactor C_{31} is a) -3 b) 8 c) 2 d) 6

Q5: If $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = 4$ evaluate the $\begin{vmatrix} 3(2a+g) & 3(2b+h) & 3(2c+i) \\ d & e & f \\ a & b & c \end{vmatrix} =$ a) -12 b) -1 c) 9 d) 8

Q6: Let $A = \begin{bmatrix} 1 & 3 & 12 \\ 0 & 2 & 6 \\ 0 & 0 & 2 \end{bmatrix}$ Then $|2A^{-1}| =$ a) 3 b) 1 c) 4 d) 2

Part B: Fill the blank

(12 marks)

Q1: The values of the unknown constant in order for $A = \begin{bmatrix} 2 & a-b & 5 \\ 3 & 4 & 6 \\ a & 6 & 3 \end{bmatrix}$ to be symmetric is $a=5$
 $b=2$

Q2: Let $A^{-1} = \begin{bmatrix} 2 & -5 & -1 \\ 3 & 1 & 2 \\ 5 & 0 & 4 \end{bmatrix}$ Then $(3A^T)^{-1} = \begin{bmatrix} \frac{2}{3} & 1 & \frac{5}{3} \\ -\frac{5}{3} & \frac{1}{3} & 0 \\ -\frac{1}{3} & \frac{2}{3} & \frac{4}{3} \end{bmatrix}$

Q3: The values of X in order for $A = \begin{bmatrix} x-4 & 4 & 6 \\ 0 & x+3 & 0 \\ 0 & 0 & x-5 \end{bmatrix}$ to be invertible is $\mathbb{R} / \{4, -3, 5\}$

Q4: The 3×3 matrix $A = [a_{ij}]$ whose entries satisfies the stated condition $a_{ij} = j - i$, $A = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 1 \\ -2 & -1 & 0 \end{bmatrix}$

Q5: Let $A = \begin{bmatrix} 1 & 5 & 6 & 2 \\ 2 & 4 & -2 & 1 \\ 6 & 12 & -6 & 3 \\ 1 & 0 & -1 & 2 \end{bmatrix}$ show that $\det(A)=0$, without directly evaluating the determinant, by state the property which use. $R_3 = 3R_2$

Q6: The values of k for which $\det(A)=0$, where $A = \begin{bmatrix} k-4 & 0 & 0 \\ 5 & k & 0 \\ 3 & 1 & k+2 \end{bmatrix}$ is $k=4, 0, -2$

Part C: Written questions

(8 marks)

Question 1: For some nonsingular square matrix A, assume that $A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & -1 & 2 \end{bmatrix}$ solve the

system $AX = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

Question 2: Find the augmented matrix for the given system of linear equation.

$$\begin{aligned} x - 3y + z &= 4 \\ y &= -9z + 7 \\ x + 3y &= 1 \end{aligned}$$

Question 3: Show that the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$ is invertible for all values of θ and find A^{-1} using adjoint.

With my best wishes _

عنوان (-)

1 $X = A^{-1} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

$X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -4 \end{bmatrix}$ 2

2

$\begin{bmatrix} 1 & -3 & 1 & 4 \\ 0 & 1 & 9 & 7 \\ 1 & 3 & 0 & 1 \end{bmatrix}$ 1

3

$A^{-1} = \frac{\text{adj } A}{|A|}$ 1

$|A| = \cos^2 \theta - \sin^2 \theta = 1$ so invertible 1

$\text{adj } A = [c_{ij}]^T = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}^T$ 2

$A^{-1} = \frac{\text{adj } A}{1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}^T$ 1